GENOMIC EPIDEMIOLOGY of dengue in THE PHILIPPINES

2022-2024

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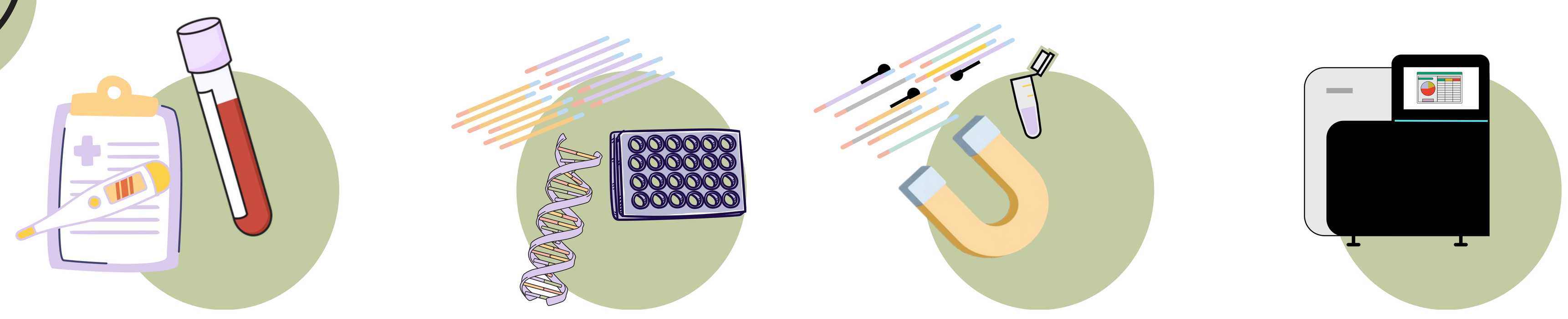
1 COSMOS: pan-arboviral sequencing



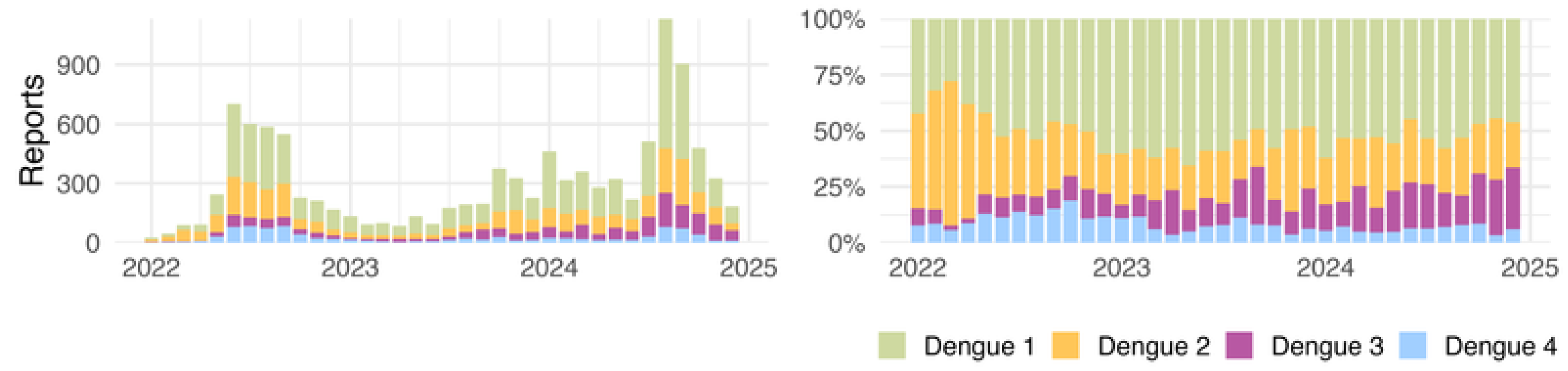
Serum
from suspected dengue cases

Metagenomics
RNA extraction and library prep

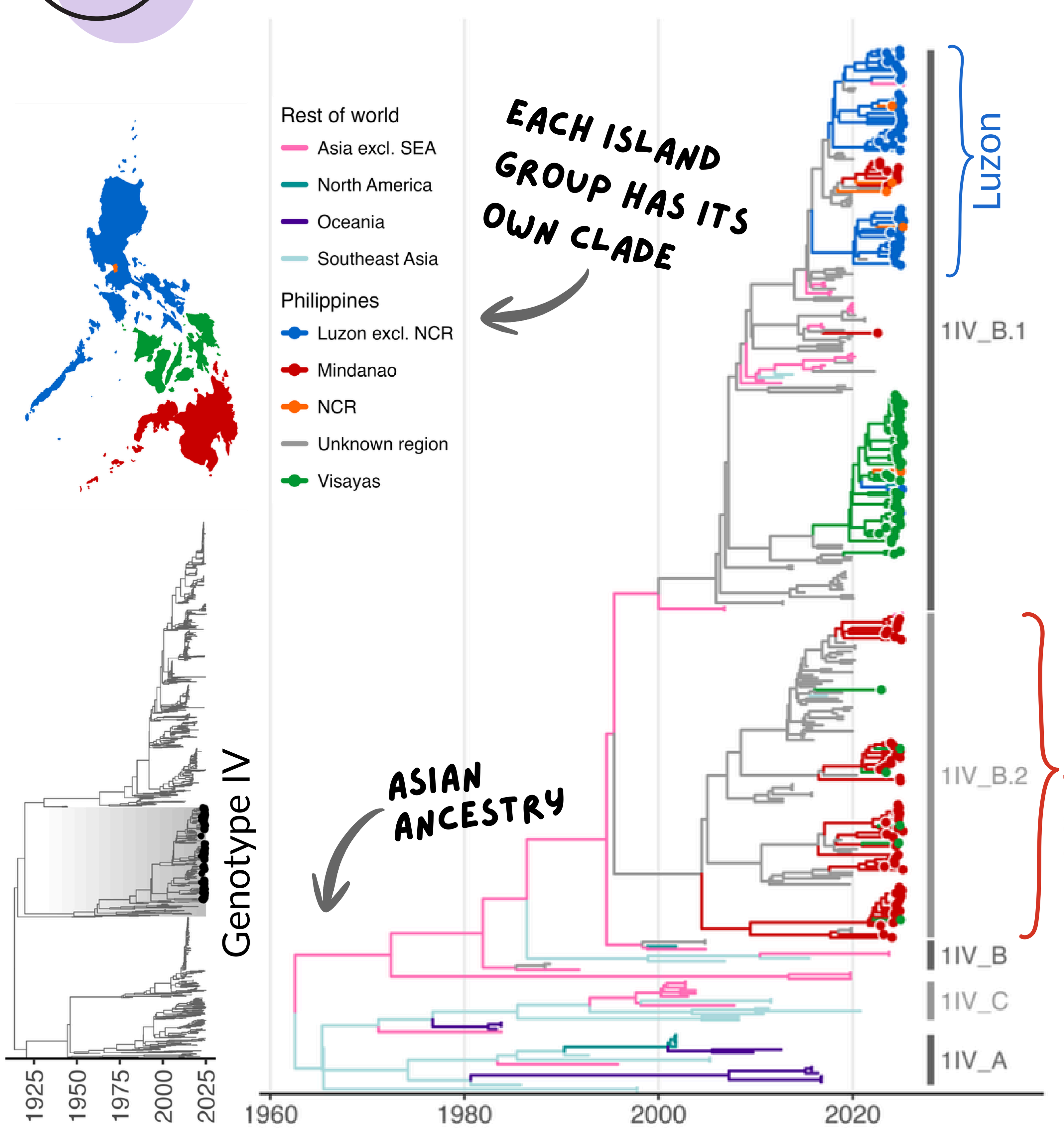
Bait panel
arboviruses, vector *cox1*, other pathogens

NGS
llumina short-read sequencing

National surveillance program: **all serotypes circulate**

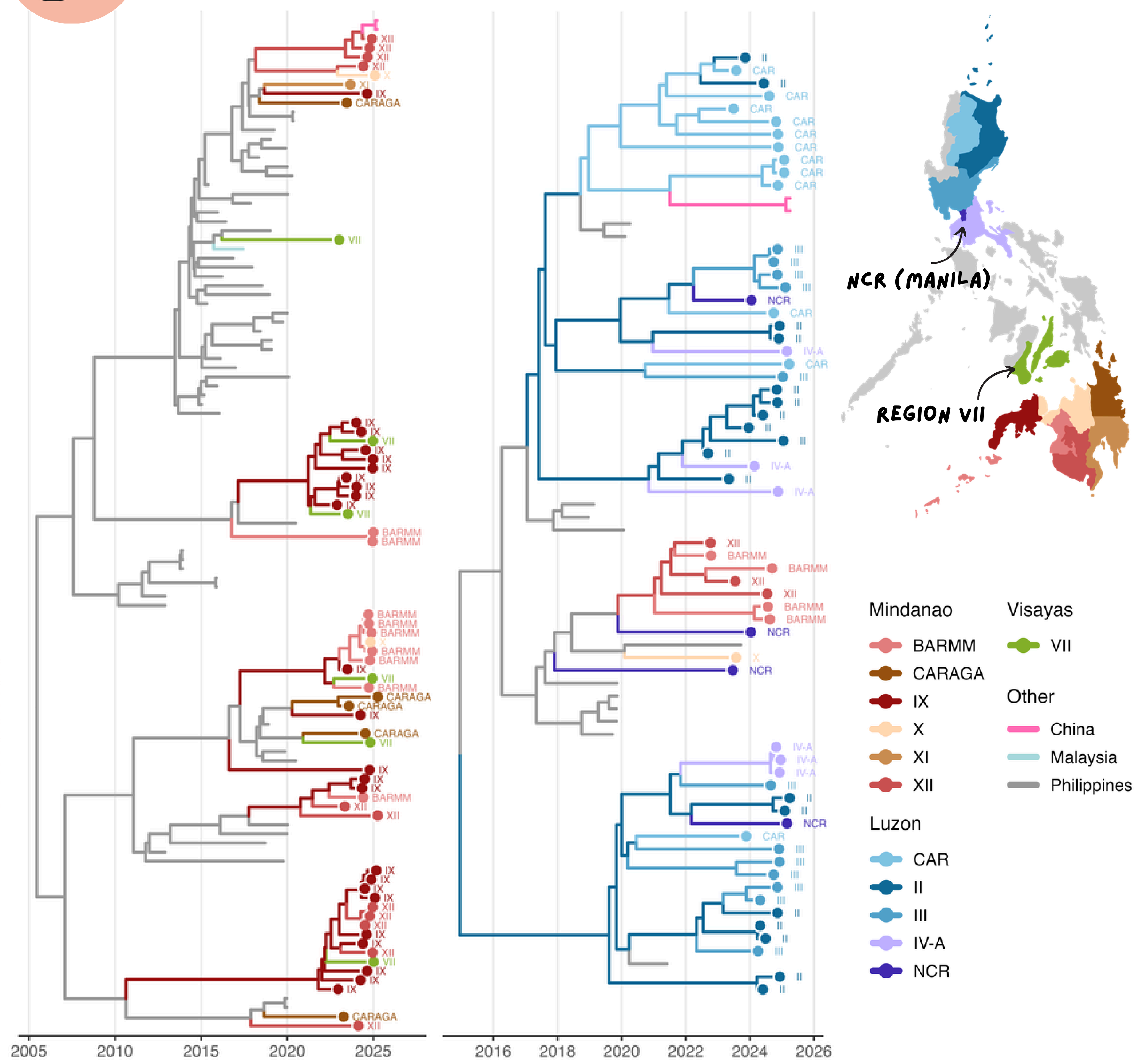


2 Endemic DENV1 circulation



Phylogeographic analyses with BEAST

3 with mixing between key regions



Mindanao (1IV_B.2) Luzon (1IV_B.1)

4 The panel includes non-DENV targets

PCR serotyping / COSMOS detection

PCR serotyping	DENV1	DENV2	DENV3	DENV4	no result	EBV	CHIKV	HHV-6	adenovirus	JC virus	parvovirus B19	BK virus	entreovirus A	S. aureus
Negative	9	3	3		287	30	6	3	1	1	2	1	1	
DENV4				17	17	1								1
DENV3			27		8									
DENV2		38			17									
DENV1	181				44	9		2	1	1				

CHIKUNGUNYA

> 100 reads
> 30% coverage

NOT FOUND

We sequenced PCR-negative samples (~half)
Detected other targets, with some co-detections.

QUALITY

DENV whole genome recovery was good
For the 2024 cohort, almost all PCR-positive have genomes.
For older samples, RNA quality lowered the success rate.

SCAN QR CODE

LINKS, PAPERS, DIGITAL VERSION

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ReSPond project: RITM + Sydney

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